### Laboratory Work № 1

- **Topic:** Conditional Statements and Decision-Making in Python
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# Objective

- Apply conditional statements (`if`, `elif`, `else`) to solve real-world problems;
- Develop decision-making algorithms;
- Practice nested conditions and logical operators;
- Improve problem-solving and programming skills through interactive applications.

# Tasks and Solutions

## Task 1. Body Mass Index (BMI) Calculator

A program that calculates a user's Body Mass Index (BMI) based on weight and height inputs and determines the corresponding health category.

Python
```
# Task 1: BMI Calculator
a = int(input("enter weight (kq)  "))
b = input("enter height (m)   ")
c = float(b)
BMI = a / (c * c)
if BMI < 18.5 or BMI == 18 / 5:
      print("underweight")
elif BMI < 24.5:
      print("normal weight")
elif BMI < 29.9:
      print("over weight")
else:
      print("obese")
```

## Task 2. Chinese Zodiac Calculator

A program that determines the Chinese Zodiac sign based on a given birth year using a 12-year cycle (`year % 12`).

Python
```
# Task 2: Chinese Zodiac Calculator
a = int(input("enter birthdate:  "))
b = a % 12
if b == 0:
  c = "monkey"
elif b == 1:
  c = "rooster"
elif b == 2:
  c = "dog"
```

```
elif b == 3:
  c = "pig"
elif b == 4:
  c = "rat"
elif b == 5:
  c = "ox"
elif b == 6:
  c = "tiger"
elif b == 7:
  c = "rabbit"
elif b == 8:
  c = "dragon"
elif b == 9:
  c = "snake"
elif b == 10:
  c = "horse"
else:
  c = "sheep"
print("your Chinese Zodiac sign is: ", c)
```

## Task 3. Birthday Guessing Game

A program that calculates the user's birthday using binary number logic by asking whether the birthday is included in five predefined sets.

Python

```
# Task 3: Birthday Guessing Game
a = 0
print("is your set is in Set 1 ? ,Enter 1 for yes and 0 for no :")
print("Set1 is :{1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29,
31}")
if input() == "1":
  a = a + 1
print("is your number is in Set 2 ? ,Enter 1 for yes and 0 for no :")
print("Set2 is :{2, 3, 6, 7, 10, 11, 14, 15, 18, 19, 22, 23, 26, 27, 30,
31}")
if input() == "1":
  a = a + 2
print("is your number is in Set 3 ? ,Enter 1 for yes and 0 for no :")
print("Set3 is :{4, 5, 6, 7, 12, 13, 14, 15, 20, 21, 22, 23, 28, 29, 30,
31}")
if input() == "1":
  a = a + 4
print("is your number is in Set 4 ? ,Enter 1 for yes and 0 for no :")
print("Set4 is :{8, 9, 10, 11, 12, 13, 14, 15, 24, 25, 26, 27, 28, 29, 30,
31}")
if input() == "1":
  a = a + 8
print("is your number is in Set 5 ? ,Enter 1 for yes and 0 for no :")
print("Set5 is :{16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29,
30, 31}")
if input() == "1":
  a = a + 16
print("your birthday is :", a)
```